Hao Lee

Seattle, WA | (206) 843-2820 | Howard.lee1121@gmail.com | haollll.github.io | LinkedIn | GitHub

Summary

Master's student at the University of Washington specializing in robot learning and control, with hands-on experience across VLA systems, reward learning, imitation learning, and real-world robot deployment.

Education

University of Washington Sep 2025 – Present
Master of Science in Mechanical Engineering (Data Science Track) *Seattle, U.S.A*
– Related coursework: CSE546 ML / CSE571 AI-Robotics / CSE579 Reinforcement Learning / EE546 Control Sys Theory

National Cheng Kung University Sep 2021 – Jun 2024
International Bachelor's Degree of Energy Engineering, Aerospace and Aeronautical Engineering *Tainan, Taiwan*

Professional Experience

Research Assistant May 2026 – Present
Robot Learning Lab, Paul G. Allen School, UW *Seattle, U.S.A*
– Built a complete sim-to-real deployment pipeline for DROID-pretrained priors on a real Franka arm, resolving proprioception, rotation representation, and policy stability issues
– Led manipulation task design and execution for full-scale real-robot experiments, including debugging and evaluation, contributing to a manuscript in preparation

Data Engineer Intern Jun 2024 – Aug 2024
AUO-Digitech *Hsinchu, Taiwan*
– Processed and curated large-scale manufacturing datasets to build training pipelines for predictive ML models; performed hyperparameter tuning and performance evaluation using PyTorch-based frameworks
– Supported cross-functional R&D on drone applications in factory automation

Research Experience

LLM-Inferred Reward Functions for Task-based RL Apr 2026 – Jun 2026
Capstone Project, UW (CSE579 Reinforcement Learning, advised by A. Gupta) *Seattle, U.S.A*
– Built an RLHF-style reward learning pipeline that replaces human annotators with an LLM: given environment source code, an LLM generates both a fixed reward function r_{fixed} and a semantic trajectory summarizer via RAG
– Trained an ensemble reward model (3× MLP, Bradley-Terry loss) with active learning via disagreement scoring to minimize LLM queries; combined reward $r_{total} = r_{fixed} + g \cdot R_\phi$ used to train PPO on HalfCheetah-v5

Vision-Language-Action (VLA) Robot Control Pipeline Jan 2026 – Mar 2026
Capstone Project, UW (CSE571 AI-Robotics, advised by Dieter Fox) *Seattle, U.S.A*
– Built an end-to-end 4-stage VLA pipeline: natural-language command → semantic waypoint planning → 3D projection → Cartesian trajectory execution on a 6-DOF robot arm
– Integrated Google Gemini as vision-language planner to infer semantic 2D waypoints from RGB frames, then lifted pixel-space waypoints into 3D robot-frame coordinates via stereo depth projection with spatial-temporal median filtering

Multi-Drone Autonomous Minefield Mapping & Path Planning Jan 2026 – May 2026
Washington Aerial Robotics Team (IARC Mission 10) *Seattle, U.S.A*
– Architected peer-to-peer coordination layer using ROS2/FastDDS with gossip-protocol-style diffusion across 4 autonomous agents, achieving sub-100ms propagation latency
– Designed decentralized task-auction mechanism enabling agents to independently estimate costs and compete via claim broadcasts and implemented hierarchical state machine with priority-ordered behavior tree for reactive collision avoidance

GNN Formation Control for Multi-Drone Systems Jan 2026 – Mar 2026
Capstone Project, UW (EE546 Advanced Control Theory) *Seattle, U.S.A*
– Designed a GNN policy trained via imitation learning to maintain chain formation across leader-follower drone networks
– Achieved zero-shot generalization from n=4 to n=8 followers using permutation-equivariant shared weights

Deep Reinforcement Learning for Quadrotor Flight Control — Undergraduate Thesis Sep 2023 – Jun 2024
NCKU MARS Lab (Dr. Chen) *Tainan, Taiwan*
– Implemented DDPG algorithm in simulation to train an autonomous quadcopter flight controller
– Performed ablation studies on reward shaping, action-space normalization, and noise injection to optimize flight policy

Technical Skills

– **Languages, Frameworks & Tools**: Python, C++, Isaac Sim, ROS2, PyTorch, SQL, MuJoCo, CUDA, Docker, Git
– **Robotics & Control**: Robot Manipulation, MPC, Path Planning, SLAM, Behavior Trees, FK/IK, Multi-Agent
– **ML / AI**: Deep RL, GNN, Transformer, World Models, Imitation Learning, Multimodal LLM, VLA